

HOUSEHOLD CONSUMPTION OF ELECTRICITY IN BRAZIL BETWEEN 1985 AND 2013

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Abstract

This article describes the electricity consumption in Brazilian residences between 1985 and 2013 through linear regressions. The explanatory variables considered were the number of households, effective consumption of families as a proxy for family income, and electricity tariff for households. To deal with the power generation crisis of 2001 we have introduced a dummy variable in the form of a step function. With such explanatory variables, we were able to account for the reduction of household electricity consumption caused by the policies conducted in 2001 and their permanent consequences. The regression presented coefficient of determination of 0.9892, and the several statistic tests conducted assured the existence of long-term relation between the electricity consumption in residences and the explanatory variables. The obtained elasticities for the household consumption of electricity with respect to number of residences, family income and residential tariff of electricity were 1.534 ± 0.095 , 0.189 ± 0.049 , and -0.230 ± 0.060 , respectively. These results allowed

understanding the evolution over time of the household consumption of electricity in Brazil. They suggest that the electric sector in Brazil should pursue an active policy to manage demand of residential electricity using tariffs as a means to control it.

Keywords: Residential electricity consumption, Brazil, household income, electricity tariff, number of households, households appliances.

1. Introduction

The electricity consumption in residences is normally explained by economic factors such as economic activity, family income, electricity tariffs and those of its substitutes, environmental conditions, increasing population, number of residential units, etc (Nakajima and Hamori, 2010; Achão and Schaeffer, 2009; Sa'ad, 2009; Ghisi et al, 2007; Narayan et al, 2007; Halvorsen and Larsen, 2001; Jannuzzi and Schipper, 1991). Other explanatory factors relate to changes in life style, public policies, climate and regional conditions, etc., that induce new behavior, habits and basic needs (Wiesmann et al, 2011; Achao and Schaeffer, 2009; EPE, 2007; Jannuzzi and Chipper, 1991; Ghisi et al, 2007; Halvorsen and Larsen, 2001). Most models reported in the literature consider family income, electricity tariffs, and other complementary factors or explanatory variables which the authors consider important to explain its behavior.

For Portugal, a model to describe the electricity consumption in residences included as complementary explanatory variables the household area, persons per household, weather conditions and many other factors (Wiesmann et al., 2011). For the G7 countries, a similar model considered as important complementary variables natural gas tariff, the ratio between electricity and natural gas tariffs, price of home appliances, and ambient temperature. The relatively high magnitudes of the elasticity with respect to electricity and natural gas tariffs suggested their potential use to control residential electricity demands (Narayan et al., 2007). For Norway, the authors considered as important complementary variables the environment temperature, price and number of electric home appliances. They identified as causes for the increase in household electricity intensity the greater use

of appliances and larger home areas, while changes of consumer habits were considered without impact on it (Halvorsen and Larsen, 2001).

For Brazil, Achão and Schaeffer (2009) identified as important complementary factors affecting the residential consumption the number of households and the household electricity intensity. Factors such as the country's regions and income levels that could be associated with different habits or behaviors had smaller impact on the residential electricity consumption (Achão and Schaeffer, 2009). Other important factors were the inclusion of low-income populations as consumers under the universalization policy of electricity consumption, and availability of easy credit conditions for the consumers with the consequent increase in their ownership of electrical appliances (Achão and Schaeffer, 2009; EPE, 2007; Jannuzzi and Chipper, 1991; Ghisi et al, 2007).

To describe the behavior of the consumption of residential electricity through explanatory variables one usually use linear regressions to estimate technical coefficients or elasticities. The choice of explanatory variables or factors are also limited by the availability of data for the analyses (Nakajima and Hamori, 2010; Sa'ad, 2009; Narayan et al, 2007; Gujarati, 2000).

This work aims at identifying the important factors or explanatory variables that influenced the consumption of residential electricity in Brazil between 1985 and 2013, and obtain elasticities from linear regressions with respect to them. To understand the recent evolution of the household consumption of electricity we construct scenarios assuming different evolution of explanatory variables. This article starts presenting the structure of the electricity consumption in the country, and the data and methods used in this study. It follows with results and discussions, with an attempt to understand the recent evolution of the residential consumption of electricity in Brazil, and finally with conclusions and policy recommendations.

2. Structure of the electricity consumption in Brazil

The total electricity consumption in Brazil is still small compared with of developed countries; the total electricity consumption was 2.94 MWh/person in 2011 while in developed countries it was between 5 MWh/person and 24 MWh/person (EPE, 2012). The residential electricity consumption is expected to increase in Brazil in the coming years to accommodate economic growth and

improvements on the level of human development of Brazilian society (EPE, 2012). In 2013 it represented 26 % of the total electricity consumed in the country with an average growth rate of 4.9 %/year between 2003 and 2012 (EPE, 2014).

The structure of electricity consumption in Brazil is presented in Fig. 1. The residential sector, responsible for 26 % of electricity consumption in 2013, has grown faster than the other sectors due to several factors including the increasing intensity of household electricity consumption through utilization of home appliances. Fig. 2 presents the main appliances responsible for electricity consumption in a typical Brazilian home in 2012. Electric showers are an important consumption appliance in the country, and its presence decreased slightly over the last 10 year (EPE, 2014). The use of air conditioning devices, including those with reverse cycle to provide cool and hot air as desired, is currently growing very fast in the country and may soon reach the third position regarding electricity consumption. These observations tend to indicate that greater family income has been an important factor to determine the electricity consumption in Brazil.

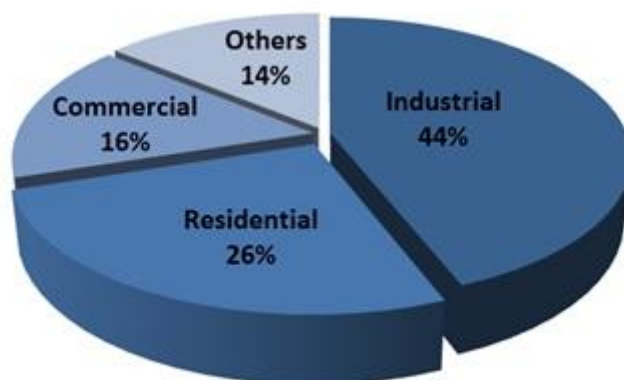


Fig. 1. Structure of the electricity consumption in Brazil in 2013 (EPE, 2014).

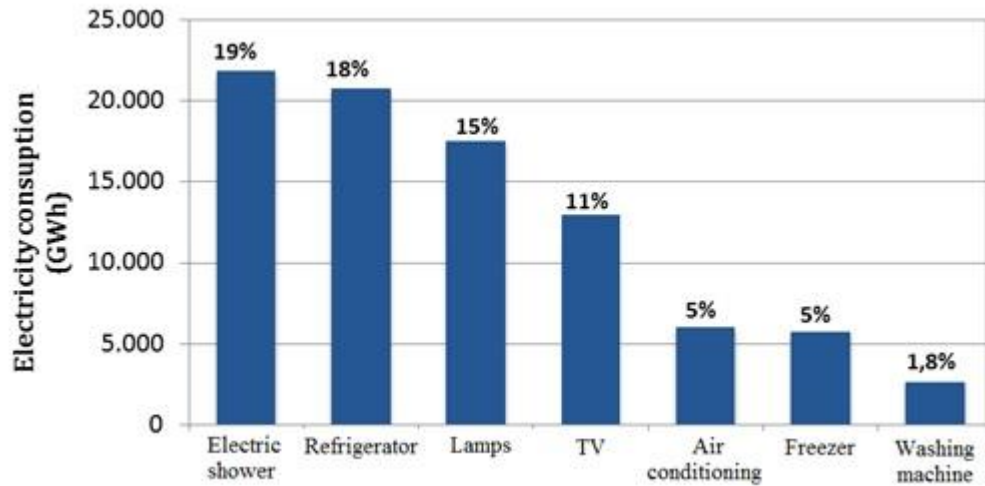


Fig. 2. Main appliances found in Brazilian households in 2012. The share of each appliance with respect to the total residential consumption is presented above each bar EPE (2012).

3. Methods and data

In this work we try to explain the household electricity consumption behavior in Brazil in a top-down manner through explanatory variables, expressed in time series, using linear regression based on the ordinary least squares approach (Gujarati, 2000). This approach furnishes elasticities relating the electricity consumption in residences to the explanatory variables. The typical variables considered in such approaches are income and tariffs. To perform the regressions and statistical tests we employed the EViews 8 software (IHS Global, 2013). Econometric theory is used to evaluate the stationarity of the time series, their order of integration and possible cointegration to verify if there are meaningful causal relations between the explanatory variables and the electricity consumption in residences.

On the other hand we use a bottom-up approach to survey specific data and factors, that may have influenced the household consumption of electricity in Brazil during 1985 and 2013, to identify a broader and more meaningful set of explanatory variables (Wiesmann et al., 2011). To do that we undertake a literature review analyzing possible factors and present it in Sect. 3.2 below. With meaningful elasticities, we discuss the results, and attempt to explain the current evolution of the residential consumption of electricity in Brazil. Specifically, we make some projections for the coming years attempting to understand the impact of the explanatory variables on it.

3.1. Regression Model

The functional form to describe electricity consumption is somewhat arbitrary. Although there is no consensus about the best functional form, many studies consider multiplicative or logarithmic models. We adopt such a model for the analysis of residential electricity consumption, E_t , i.e.,

$$E_t = b_0 \prod_{i=1}^N X_{it}^{\varepsilon_i} e^{u_t} \quad (1)$$

where X_{it} is the explanatory variable at time t , b_0 and ε_i , $i=1, \dots, N$, are coefficients to be determined, and u_t is a residue. The coefficients ε_i are interpreted as elasticities and defined as

$$\varepsilon_i = \frac{\frac{\Delta E_t}{E_t}}{\frac{\Delta X_{it}}{X_{it}}}, i = 1, \dots, N. \quad (2)$$

Eq. 1 can be treated as a linear combination by taking the logarithm on both sides of the equality, i.e.,

$$\log(E_t) = \log(b_0) + \sum_{i=1}^N \varepsilon_i \log(X_{it}) + u_t. \quad (3)$$

In this way, the coefficients, ε_i , $i = 1, \dots, N$, can be obtained through a linear regression of the logarithm of the several time series of the explanatory variables (Gujarati, 2000).

3.2. Explanatory variables for the household consumption of electricity

Studies on residential electricity consumption consider different explanatory variables (Achão and Schaeffer, 2009; EPE, 2007; Ghisi et al., 2007; Halvorsen and Larsen, 2001; Jannuzzi and Schipper, 1991; Wiesmann et al., 2011). We reviewed this literature in a bottom-up approach, and present in Table 1 the main variables identified as with potential to explain the electricity consumption in residences. The variables are divided in important and less important according to their explanatory power. In the “comment” column we summarize how these variables affect the consumption of residential electricity according to the studied literature. In a gross manner they encompass issues

related to economics, life style, public policies, regional or climate impacts, and may be represented through time series or qualitative information.

The variables adopted in this work to explain the Brazilian consumption of electricity in residences are the ones considered important by several authors for long-range analysis, and are highlighted accordingly in Table 1. They are the number of households in the country, the available family income, the electricity tariffs for residences, ownership of home appliances, and social or economic policies that directly affected the household electricity consumption. The first three variables constitute time series and are available in different Brazilian agencies and consolidated at the Instituto de Pesquisa Econômica Aplicada (IPEA) (IPEA, 2014a, 2014b, 2014c, 2014c). The last two ones affect the consumption of electricity in residences in qualitative terms and are not time series (Achão and Schaeffer, 2009; Halvorsen and Larsen, 2001; Nakajima and Hamori, 2010).

Table 1 describes explanatory possibilities of the denoted less important variables. They deal more with short term impacts or trends, and with structural relative impacts of groups of consumers, specific appliances, climatic regions, temperature, etc. They tend to explain relative differences rather than long term evolutions. For example, different income groups and life style can help explaining short term trends of specific groups of families regarding their consumption of electricity or buying specific home appliances. Regional and bioclimatic zones and ambient temperatures can explain details of different levels of consumption of electricity in different regions of Brazil, relative importance of air-conditioners in certain regions, but not its evolution over time. This research did not identify any deregulation of the electricity sector that caused relevant impact on residential consumption of electricity in Brazil, but specific only public policies which were discussed in Sect. 3.2. It did not identify either important any energy substitute for electricity long-term changes. As a final comment, the explanatory variables denoted as not important in Table 1 are, in general, more adequate to undertake a qualitative rather than statistical analysis.

Each new built residence requires power for lights, operating different home appliances, etc. Thus the number of residences is considered to pursue a strong positive correlation with the residence

consumption of electricity (Halvorsen and Larsen, 2001); in Brazil this variable has presented a growth rate between 2 and 4 %/year from 1985 to 2013 (IPEA, 2014c). The electricity tariff for households is another variable considered important to explain variations of electricity consumption in residences. The usual objective of several studies regarding this variable relates to obtaining the magnitude of its elasticity (Wiesmann et al, 2011; Halvorsen and Larsen, 2001).

Regarding the available family income, another important variable to the problem in question, there are few alternatives to represent it: the per capita gross national product (Irffi et al., 2009; Mattos and Lima, 2005; Viana and Silva, 2014), the total income from families or persons (Dilaver and Hunt, 2011; Halvorsen and Larsen, 2001; Nakajima and Hamori, 2010; Narayan et al., 2007; Wiesmann et al., 2011), or the effective consumption of families accounting for durable goods, housing, etc. (Haughton and Khandker, 2009). Important points to be considered are permanent and variable incomes, and the fact that the consumption varies with the level of family income. The per capita gross national product does not represent actual income of families and may be considered the poorest candidate of the three variables mentioned above. The total income of families takes into account the part devoted to consumption and also the one saved aiming at capital accumulation, which may be substantial but not uniform among all families (Bacha, 2004). Its variation over time may not correlate well with the consumption of electricity by households (see the entry “Income level of families” in Table 1).

The third alternative, namely, the effective consumption of families accounting for goods, housing, durable goods such as home appliances, etc, appears as a better choice to represent the family income available to consume electricity. Moreover, it takes into account policies that indirectly augment the available income for consumption of low-income families, and thus allow families to pay higher electricity bills, and increase the ownership of home appliances (see the corresponding entry in Table 1). During the period of study, an important public policy to increment the ownership of home appliances was cheap credit to low income families whose effect is also captured by the effective for purchasing home appliances (Achão and Schaeffer, 2009; EPE, 2007). Thus in his work we chose the effective consumption of goods and services by families as the explanatory variable to represent the

family income available to consumption (IPEA, 2014d). This variable will be denominated in this article as family income related to consumption or simply family income.

Table 1. Possible variables identified as having explanatory potential for the residential electricity consumption in Brazil. They are grouped with respect to their explanatory power.

Fig. 3 presents the temporal behavior of these variables obtained from IPEA namely the electricity consumption in residences (E_t), the number of households in the country (H_t), the effective consumption by families as a proxy for the available family income for consumption (I_t), and the electricity tariff for residences (T_t). All series were normalized to 1 in 1985 which is taken as a reference year. The tariff and family income variables were corrected by the annual inflation rates in order to produce consistent evaluations over time.

The electricity consumption has been increasing during the period of study except by the important drop during the power generation crisis of 2001. The number of households has increased continuously with a slight decline in the growth rate after 2005. This continuous increase in the number of residences in Brazil is thought to be an important contributor to the total consumption of residential electricity as each new homes require a minimum electricity load to meet the demand of a typical family (Achão and Schaeffer, 2009; EPE, 2007).

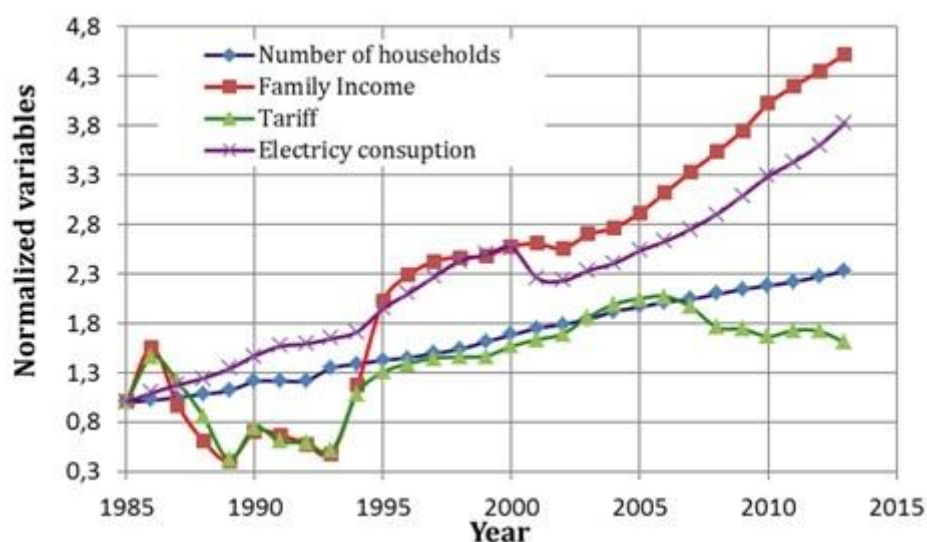


Fig. 3. Time series for residential electricity consumption, number of households, family income (consumption of families) and electricity tariff for households between 1985 and 2013 (IPEA, 2014a, 2014b, 2014c, 2014d).

The family income and electricity tariff, as monetary variable, have a complex behavior during 1985 and 1995 because of the country's high inflation in this period. Successive stabilization plans were carried out until in 1994/1995 when the currency was stabilized. Such attempts to control the high inflation rate caused the ups and downs between 1985 and 1994 (Bacha, 2004; IPEA, 2014b, 2014c). After 1995 the behavior of both variables are more predictable. After 2002 the family income presents a continuous growth. The tariff for residential electricity increases between 1995 and 2006. The decreasing behavior of the tariff after 2006 is due to price adjustments below the inflation rate during that period.

Fig. 4 presents the growth rates from 1995 to 2013 for the variables presented in Fig. 2. Note that after 2005 the growth rate for the electricity tariff for residences was negative for most of the years reaching -10 %/year in 2008. In the same period the number of households increased with rates between 2 and 5 %/year. The family income proxy adopted in this work increased with rates between 4 and 7 %/year. As a matter of comparison, the country's gross domestic product increased after 2010 at

rates between 1 and 3 %/year (IPEA, 2014a, 2014b, 2014c, 2014d). Before 1995, the growth rate of the family income and electricity tariff varied a lot due to high inflation rates and several attempts to control it (Bacha, 2004).

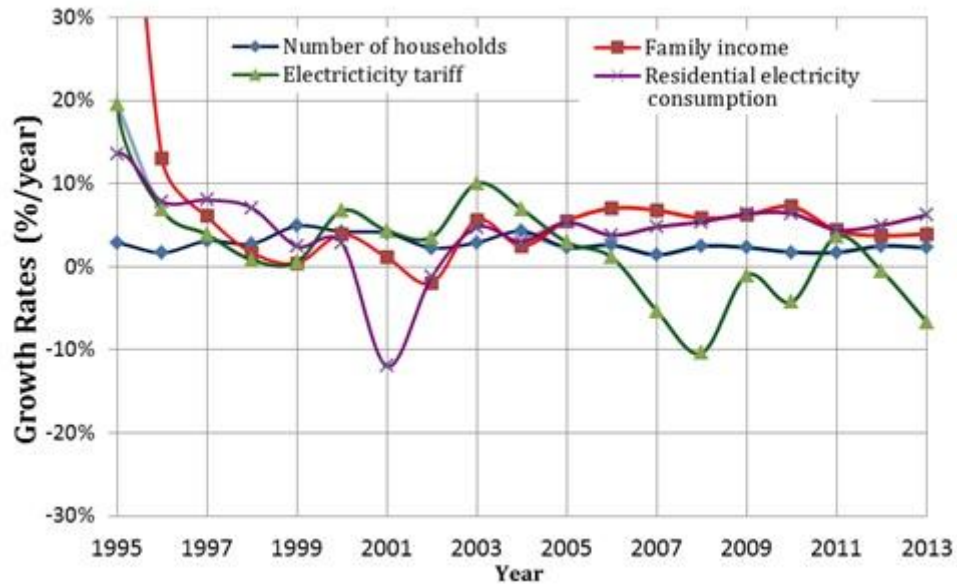


Fig. 4. Growth rates as a function of the year for the number of households, family income, electricity tariff and residential electricity consumption between 1985 and 2013.

3.3. Explanatory variables for public policies

Table 1 indicates that the qualitative variables related to the ownership of home appliances and social and economic policies may have important explanatory power. The bottom-up analysis revealed that some specific public policies may have had impact on the electricity consumption in residences in the period studied. We identify the following social policies as deserving attention to the problem in question: restraining the electricity consumption in 2001, increasing the access of low income families to electricity, and implementing energy efficiency programs (Achão and Schaeffer, 2009; Brasil, 2016; Nogueira et al., 2015).

The power generation crisis of 2001 caused a sudden drop in the electricity consumption in residences because of a specific public policy requesting from each household a consumption restraint of 20 % to cope with the drought in that year. An interesting consequence of that public policy was changing permanently the behavior of the Brazilian families regarding consumption of electricity. As shown in Fig. 2, it decreased in 2001, and did not return to the previous level in the next year when the drought and the restrain policy of electricity consumption finished. The families acquired habits of low consumption and kept the reduction during the following years, which only returned to the level of 2000 in 2006, as shown in Fig. 6 (Achão and Schaeffer, 2009; EPE, 2014).

To model the effects of the 2001 events and that permanent behavior change of the families we defined the dummy variable, D_t , as

$$D_t = \begin{cases} 1 & \text{for } t < 2001 \\ e & \text{for } t \geq 2001 \end{cases} \quad (4)$$

where e is 2.7182818... . Eq. 4, a step function, tries to model the effects of the public policy introduced in 2001 which caused a sudden change in 2001 which continued permanently in the following years. This sudden change is also evidenced in Fig. 4 which shows a steep negative peak in 2001 for the growth rate of the residential electricity consumption. Such a choice of D_t is convenient because its logarithm in Eq. 3 is easily handled since

$$\log(D_t) = \begin{cases} 0 & \text{for } t < 2001 \\ 1 & \text{for } t \geq 2001 \end{cases} \cdot \quad (5)$$

The other two public policies have opposite effect on the household consumption of electricity but are similar in their nature and implementation. Both have been established in state agencies which received funds over several years in order to carry out their goals (Brasil, 2016; MME, 2011). In 2000 about 3 million low income Brazilian families did not have access to electricity in their residences. A public policy named “Luz para Todos”, aiming at universalizing access to electricity, began in 2003 (Brasil, 2016). In 2015 3.2 million families had already started to use electricity in their residences representing additional 4.8 % of the total number of residences. In the opposite direction regarding consumption of electricity, the national program for electric energy conservation, named Procel, began its activities in 1984 with limited success. In 2001 it adopted minimum energy performance standards for several electric appliances improving its results (Nogueira et al., 2015). Around 2010 its

contribution to electric energy savings represented a reduction of about 1 % per year of the total electricity consumption in the country (MME, 2011).

One way to account for both effects simultaneously is to consider as explanatory variable the cumulative investment over the years assigned to them. To model it we define a dummy linear variable, F_t , to represent the cumulative investment over time allocated to carry out both public policies, i.e.,

$$F_t = \begin{cases} 1 & \text{for } t < 2003 \\ t & \text{for } t \geq 2003 \end{cases} , \quad (6)$$

or

$$\log(F_t) = \begin{cases} 0 & \text{for } t < 2003 \\ \log(t) & \text{for } t \geq 2003 \end{cases} . \quad (7)$$

We take as initial time for both public policies the year of 2003 assuming that at this year both policies were already presenting effective results. A possible evidence for this assumption could be variations in the growth rate of the residential electricity consumption after 2003 shown in Fig. 4. We note, however, that the resulting elasticity for this explanatory variable will represent an average value for the period, and can be positive or negative depending of the relative importance of each program. The “Luz para Todos” universalization program should yield a positive contribution the electricity consumption in residences, while the Procel efficiency program, a negative contribution.

3.4. Models to explain the household consumption of electricity

We developed three different models to represent the household consumption of electricity. In the first, model A, we considered only the variables represented by time series, i.e., number of households, the available family income, and the electricity tariff. In the second model, denoted by model B, we tried to deal with the 2001 event through a dummy variable to describe it. In the third model, denoted by model C, we included also the dummy variable to describe the “Luz para Todos” and Procel public policies.

Thus the regression presented in Eq. 3 can be written for model C as

$$\log(E_t) = \log(b_0) + \varepsilon_H \log(H_t) + \varepsilon_I \log(I_t) + \varepsilon_T \log(T_t) + \varepsilon_D \log(D_t) + \varepsilon_F \log(F_t) + u_t, \quad (8)$$

where H_t , I_t , T_t , D_t and F_t are the explanatory variables number of households, family income available to consumption, electricity tariff, the dummy variable defined by Eq. 4, and the dummy variable defined by Eq. 6. ε_H , ε_I , ε_T , ε_D and ε_F , are the elasticities of electricity consumption in residences with respect to H_t , I_t , T_t , D_t and F_t . For model A, the regression representing it is similar to Eq. 8 but taking the coefficient ε_D and $\varepsilon_F = 0$, and for model B, taking $\varepsilon_F = 0$.

4. Results and discussions

4.1. Statistic tests on the several explaining variables

Before conducting the linear regressions calculations using the ordinary least square method we performed several statistic tests to verify if it would produce meaningful elasticities and estimates for the electricity consumption in residences. The Dickey-Fuller Expanded test results applied for each time series considered in Eq. 6 indicated the existence of a unit root, i.e., the series are not stationary. However, non-stationary variables integrated in the same order may have a balanced relation in the long term (Gujarati, 2000). The results from the statistic tests performed in the explanatory and dependent variables are presented in Appendix A. From the results of stationarity analysis, using the Dickey-Fuller and Engler and Granger tests, we concluded that the series are cointegrated and that there is a real relationship between the explanatory variables and the residential consumption of electricity (Engler and Granger 1987; Gujarati, 2000).

These results show that the obtained regression produces meaningful estimates for the electricity consumption in residences. Moreover, they can be utilized for long term analysis such as the studied period (1985-2013).

4.2. Regression results

We employed Eq. 6 to describe the electricity consumption in residences in Brazil, and performed least square regressions for both models A and B with the EViews 8 software. Table 2 presents the linear regressions results and the respective R^2 for models A, B and C, and Fig. 5

compares these regressions with the actual data for electricity consumption. In Appendix A we present more details about the statistic tests of both regressions. The very low probabilities regarding the explanatory power of the variables indicate that the coefficients of models A and B are significant and both regressions have explanatory power regarding the dependent variable, $\log(E_t)$ (Gujarati, 2000; IHS Global, 2013; Engle and Granger, 1987). With regard to model C, we note that this probability for F_t is 0.5, indicating that it may not have explanatory power for $\log(E_t)$. We will discuss this point with more detail in the next section. Observing Fig. 5 and comparing the statistics of regressions A, B and C, we note that models B and C produce better fittings.

Table 2. Regression results obtained with models A and B presenting elasticities and statistic tests.

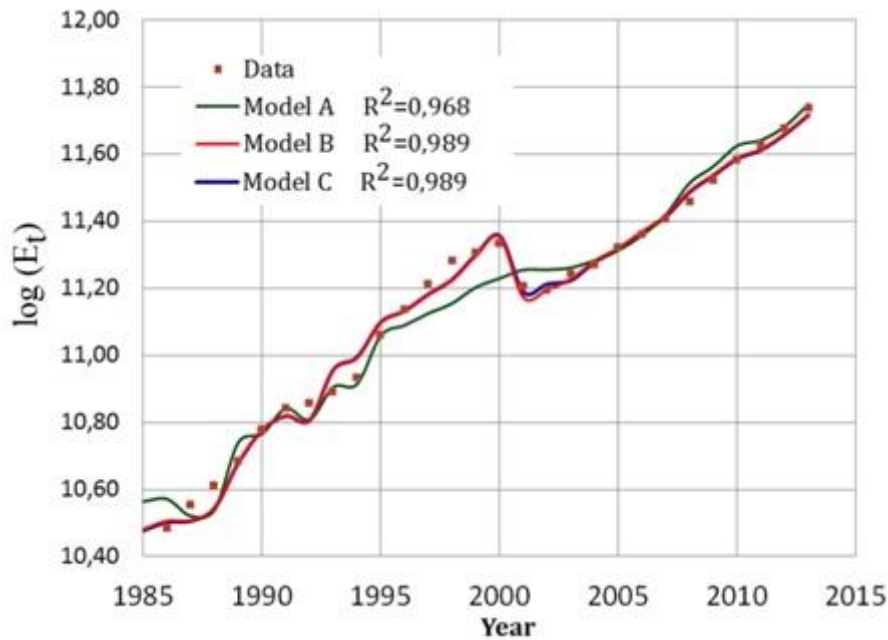


Fig. 5. Comparison between the data and regression models A, B and C representing the electricity consumption in residences during 1985 and 2013. Models B and C are almost coincident.

4.3. Discussion of the regression results

We start our discussion considering the general features of the regression presented in Fig. 5 and in the Tables of Appendix A. The signs of the elasticities agree with the premises and assumptions with respect to the variables number of households, family income and tariff. They are positive for the first two and negative for the last one. The sign of the elasticity with respect D_t is also in agreement with the premises considered. Since the dummy variable D_t presents a positive step in 2001, the drop in E_t after that year has to be accounted for with a negative elasticity. Additionally, the probabilities that these variables have no explanatory power to the problem are negligible (see Tables A1 to A3).

Regarding the dummy variable F_t , its elasticity presents an uncertainty greater than ε_F itself (see Table 2), and a 50 % probability that F_t may not have explanatory power to describe the consumption of electricity in residences (see Table A3). The first observation allows us to conclude that its sign is not meaningful and may be positive or negative, and the second, that F_t may be a poor choice as explanatory variable. On the other hand, if we consider F_t as a meaningful explanatory variable we note that the negative sign of its elasticity indicates that the Procel energy efficiency program had a greater impact on the consumption of electricity in residences than the “Luz para Todos” electricity universalization program during 2003 and 2013. When we compare the magnitude of the several elasticities, we note that ε_F is much smaller than ε_H , ε_T , ε_I , and ε_D , about 1,3 % of ε_H and 10 % of the other three elasticities. This observation indicates that either one public policy cancels the effect of the other or both have a minor impact on the total consumption of electricity in Brazilian residences. The latter statement is supported in the literature (MME, 2011; Nogueira et al., 2015; Brasil, 2016) but it does not decrease the importance of these public policies which present other social and economic merits.

Since models B and C present almost coincident results and the explanatory power of F_t is not high, we choose model B as the best regression to continue our analysis. Thus the consumption of electricity in Brazilian residences is described as

$$\log(E_t) = -15,91 + 1,53 \log(H_t) + 0,188 \log(I_t) - 0,230 \log(T_t) - 0,233 \log(D_t). \quad (9)$$

Considering Eq. 2 and the results of model B we have that a rise of 1 % in the number of residences increases the electricity consumption by 1.53 %, while a rise of 1 % in the family income increases the electricity consumption by only 0.19 %. On the other hand, a rise of 1 % in the electricity tariff decreases the electricity consumption by 0.23 %.

The elasticities for models A and B show differences. Model B regression, which includes the dummy variables to represent the event of 2001, presents higher elasticity of electricity consumption with respect to the number of households and lower elasticities with respect to tariff and family income. Thus the effect of introducing a dummy variable in model B was to increase the explanatory importance of the variable number of households and to decrease the explanatory importance of the variables family income and residential tariff.

The greater value of the elasticity, ε_H , in Eq. 9 shows that electricity consumption is more sensitive to changes in the number of households than to changes in the family income and tariffs. The latter have elasticities of close absolute values meaning that their impacts on the consumption of electricity are similar. The impact of variations in the family income can be compensated by similar variations in the tariffs since they have opposite signals.

Fig. 5 shows that the dummy variable D_t was able to explain the events of 2001. The actual residential electricity consumption reduced by 12 % from 2000 to 2001. Model A, without the dummy variable, produces an estimation curve that does not capture the sudden drop in 2001. Its larger errors occur around that year, between in 1997 and 2003. On the other hand, Model B, covering the period from 1985 to 2013, was able to capture it and estimated that drop as 16.5 %.

The results of Table 2 are in general agreement with others presented in the literature which attempt to describe the electricity consumption in residences with different explanatory variables. Developed countries present elasticities of electricity consumption in households with respect to different proxies of the family income varying from 0.10 to 0.52, and with respect to residential tariffs, from -0.13 to -1.56 (Halvorsen and Larsen, 2001; Narayan et al., 2007; Sa'ad, 2009; Wiesmann et al., 2001; Viana et al., 2014. The regressions of such studies included as explanatory variables

proxies of the family income, electricity tariffs, and other variables considered important to explain the residential consumption of electricity. The results of this study lie in the lower part of these intervals.

Sa'ad (2009), considering as explanatory variables only the GNP per capita and the tariff, obtained for South Korea a higher elasticity with respect to GNP per capita, 1.33, and with respect to electricity tariff, -0.35. A similar regression was performed for the Brazilian data considering only family income and electricity tariff as explanatory variables. The obtained elasticities were 0.85 and -0.91 with respect to family income and electricity tariff, respectively, with coefficient of determination of 0.86. As was mentioned above when we compared models A, B and C the absence of explanatory variables increased the importance of those kept in the models. This observation indicates that the elasticity values of income and tariff are model dependent. The model should be able to capture the specific characteristics of the residential electricity sector under study.

4.4. Understanding the current residential consumption of electricity in Brazil

Between 2013 and 2015 the power generation sector in Brazil, heavily based on hydropower, has been under stress due to low rainfalls and meeting the power generation demand became an important issue (Moreira et al., 2015). The overall electricity demand in the country continued to grow and, to complicate matters, the electricity consumption in residences has been growing since 2005 at rates between 3.5 %/year and 6 %/year (see Fig. 3). This growth has frequently been acknowledged to the rise of family income in recent years, but the model B evidenced the importance of falling residential tariffs.

In Table 3 we present different scenarios defined in terms of growth rate of the explanatory variables of model B. Scenarios S1 and S2 are in agreement with those observed in previous years, since they present similar trends to that observed between 2005 and 2013. Scenarios S3 and S4 consider high tariff rise and two rates for the family income (moderate and high). Scenarios S5 and S6 consider very high tariff rise and similar rates to S3 and S4 for the family income. Scenario S7 attempts to consider extreme conditions (very high tariff rise and strong reduction in family income). In 2014 and 2015 the tariffs for household consumption of electricity presented increases as high as 50

%, and important family income reduction. The growth rate considered for the number of households, 2 %/year, is around its historical value.

Table 3. Scenarios for projections considering the combined effects of the explanatory variables.

Figure 6 shows the behavior of the electricity consumption for the seven scenarios set out in Table 3. For scenarios S1 and S2, with reduction in electricity tariffs, the electricity consumption in residences increases as much as 10 % in two years and follows the trend of the previous years. For scenarios S3 and S4, with high electricity tariff rise, the electricity consumption in residences increases about 3 %. For scenarios S5 and S6, with very high tariff rise, the residential consumption falls about 5 % in two years. For the S7 scenario, with very high tariff rise and strong reduction in family income, the residential consumption falls about 12 % in two years. Scenarios S5 and S7 may represent limiting scenarios of the Brazilian situation in 2015. They show important reduction in electricity consumption in residences and evidence the impact of tariffs on its evolution.

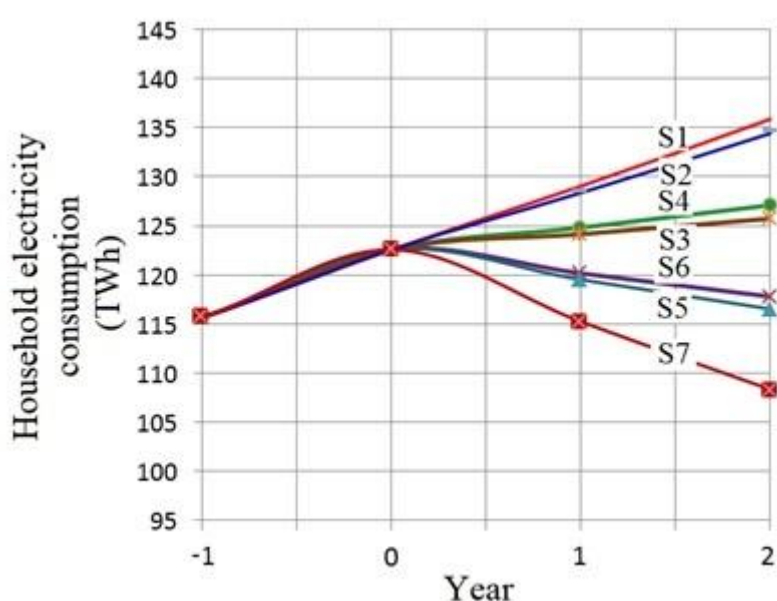


Fig. 6. Projections of electricity consumption in residences for different scenarios for the evolution of the explanatory variables.

We note in Fig. 6 that family income rise contributes to increase consumption of electricity in scenarios S1 to S4 probably due to increasing ownership of home appliances. One possible way to reduce the electricity consumption in residences under this circumstance is to introduce public policies aiming at this objective focused on specific home appliances. Two of them deserve attention, namely electric showers (see Fig. 2), and air-conditioners (EPE, 2012). Regarding electric showers, it would be interesting to discourage their use through heavy taxes, improve their energy efficiency, and facilitating the penetration of solar heated showers and gas-fired showers in Brazilian residences. The ownership of air-conditioners in the country has been growing very quickly in recent years. To reduce their impact on electricity consumption in residences we could aim at improving their energy efficiency. For both electric showers and air-conditioners, it would be interesting to review efficiency limits, improve efficiency monitoring, and continuously impose stricter efficiency regulations (Nogueira et al., 2015).

Currently the Brazilian agency for electric energy introduced a tariff flag system to compensate for greater or smaller electricity generation by high-cost fossil fuel power plants (ANEEL, 2016). In average, the tariffs can be reduced or increased from 3 to 10 % according to the indicated flag. An important question that arises is how electricity consumption by households may vary with such tariff changes. Despite the fact that the elasticity ε_T is a long-run parameter, it can be used to estimate such an impact. For example, a strong tariff reduction of about 10 % may increase the household consumption of electricity by 2.3 % during the course of one year. Conversely, a tariff increase of 10 % may reduce the electricity consumption by 2.3 % in one year. Such variations are not negligible and must be considered by energy planners.

Additionally, these results indicate that tariffs may be considered a means to manage residential consumption of electricity in the country. Although the elasticity of residential electricity consumption with respect to tariff is not high in Brazil, it is sufficient to control the electricity demand in residences

or even reduce it, as shown in scenarios S5 to S7. If the Brazilian agencies had pursued an active behavior to avoid the tariff reduction shown in Figs. 3 and 4, the electricity consumption in the residential sector would not have increased as it did and the stress in the power generation sector would have been lower.

5. Conclusions and policy recommendations

This study attempted to describe the electricity consumption in the residential sector in Brazil through explanatory variables aiming at obtaining long-range elasticities. The period of study was between 1985 and 2013, the explanatory variables considered were the number of households, effective consumption of families as a proxy for family income, and electricity tariff for households. Through the effective consumption of families, we expected to incorporate information regarding the consumption of electricity by families, but also the ownership of home appliances, and credit availability that facilitated their purchase. We have utilized dummy variables to model the power generation crisis of 2001, the “Luz para Todos” program to increase access to electricity, and the Procel energy efficiency program. With such explanatory variables, we were able to account for the reduction of household electricity consumption caused by the policies conducted in 2001 and their permanent consequences. The regression presented coefficient of determination, R^2 , of 0.989, and the several statistic tests conducted assured the existence of long-term relation between the electricity consumption in residences and the explanatory variables. The public policies adopted in the country to increase access to electricity and to increase electric energy efficiency had a minor effect on the total electricity consumption in Brazilian residences.

The obtained elasticities for the household consumption of electricity with respect to number of residences, family income and residential tariff of electricity were 1.534 ± 0.095 , 0.189 ± 0.049 , and -0.230 ± 0.060 . According to these results, the evolution of the household electricity consumption over time is more sensitive to the evolution of the number of households in the country, followed by those of tariffs and family income which have similar elasticities but with opposite signs.

The strong growth of electricity consumption in residences in Brazil has been acknowledged to the rise of family income in recent years. The values of the obtained elasticities and scenario studies conducted in this work showed that other factors played important roles on its evolution, namely the evolution over time of the number of households in the country and the electricity tariff for residences. The first presents a historical growth rate around 2 %/year and high positive elasticity. The tariffs, with negative elasticity, decreased between 2005 and 2013 at rates around 5 %/year because of price adjustments below the inflation rate. Thus both variables also contributed to the high electricity consumption in the Brazilian residential sector in recent years.

These results allow some policy recommendations regarding managing the demand of electricity in Brazilian residences. The first is that the Brazilian agencies responsible to manage the electric sector in the country should make tariff adjustments in accordance with the inflation rate since falling real tariffs does contribute to increase the consumption of residential electricity. The second is pursuing an active policy to manage demand of residential electricity and tariffs may be utilized as a means to control residential consumption of electricity. Although the elasticity of residential electricity consumption with respect to tariff is not high in Brazil, this study shows that it is sufficient to control the electricity demand in residences. As a future work, we intend to study how to implement such a policy considering the Brazilian regulation framework.

Appendix A

The statistical verifications performed in this work were the Dickey-Fuller Expanded, Engle and Granger and Durbin-Watson tests (IHS Global, 2013; Gujarati, 2000; Engle and Granger, 1987). The Dickey-Fuller Expanded test was used to evaluate the stationarity of the time series and their order of integration. The Engle and Granger test was applied to the residual series to identify cointegration and presence of long-run equilibrium among the time series. The Durbin-Watson test was used to verify the stationarity and cointegration of the residues from the obtained regressions.

Tables A1 to A4 show the results and statistics related to the regression models A, B and C obtained with the EViews 8 software for representing the evolution of the household electricity consumption in Brazil. Fig. A1 shows the residues from the residential consumption of electricity estimated by Model B. More details about these results are found elsewhere (IHS Global, 2013; Gujarati, 2000).

Table A1. Results of regression and statistic tests for Model A.

Table A2. Results of regression and statistic tests for Model B.

Table A3. Results of regression and statistic tests for Model C.

Table A4. Expanded Dickey-Fuller test applied to the series of residuals for Model B.

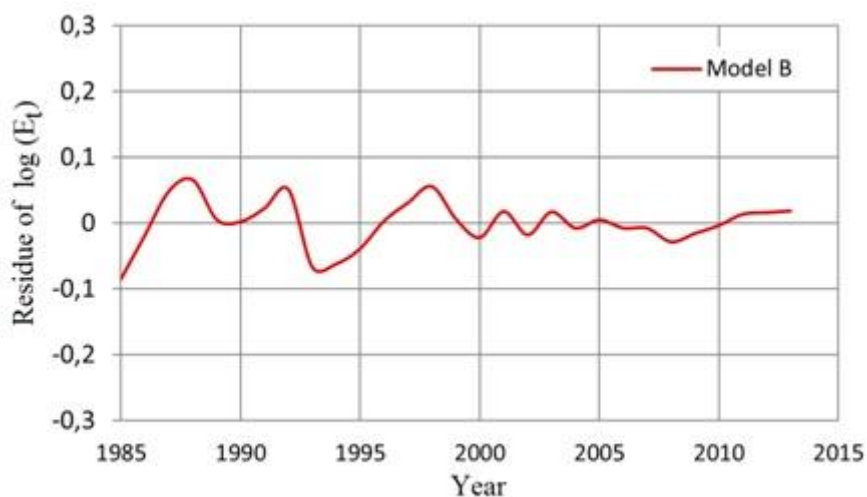


Fig. A1. Residues from the residential consumption of electricity estimated by Model B.

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